

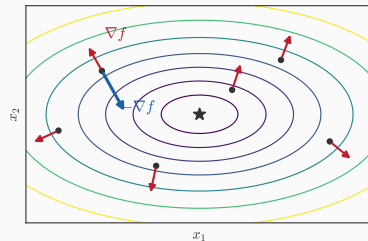
Optimization

3.3.1 Steepest descent method

José Firmino Aguilar Madeira

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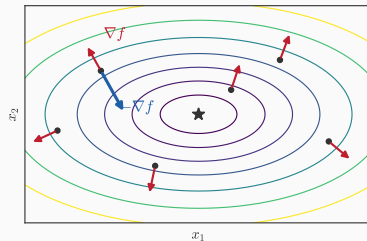


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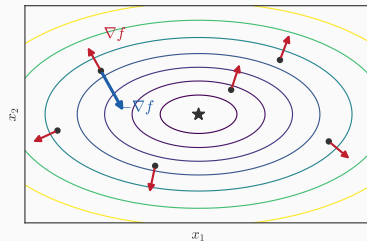
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Steepest descent = the steepest direction **at the point**, for fixed $\|\mathbf{S}\|$ — not “the algorithm that reaches the minimum fastest”.



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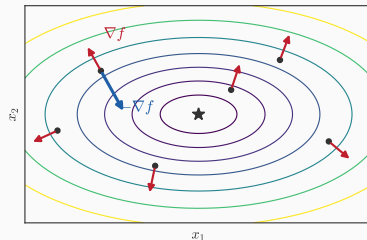
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And **how far to go**? That is a separate decision: the optimal step length λ_k^* , as in the line search of Ch. 2.

3.3 chooses $\mathbf{S}_k = -\nabla f(\mathbf{x}_k)$

Ch. 2 chooses λ_k^*



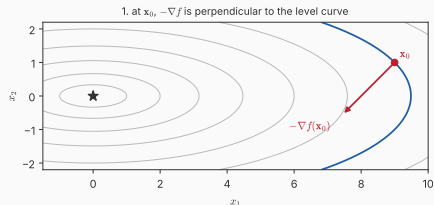
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One iteration by hand — and orthogonality appears

$f(\mathbf{x}) = \frac{1}{2}(x_1^2 + 9x_2^2)$, $\nabla f = (x_1; 9x_2)$, minimum at $(0, 0)$.

Step 1 $\mathbf{x}_0 = (9; 1)$, $f(\mathbf{x}_0) = 45$.

Step 2 $\nabla f_0 = (9; 9)$; set $\mathbf{S}_0 = -\nabla f_0 = (-9; -9)$.



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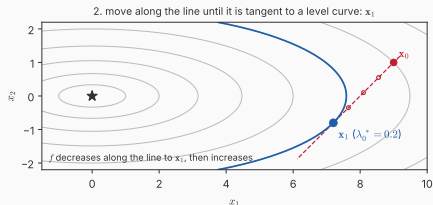
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Step 3 $f(\mathbf{x}_0 + \lambda \mathbf{S}_0) = \frac{1}{2}[(9 - 9\lambda)^2 + 9(1 - 9\lambda)^2]$;

$$\frac{df}{d\lambda} = 810\lambda - 162 = 0 \Rightarrow \lambda_0^* = 0.2.$$

Check (2nd method, quadratics):

$$\lambda_0^* = \frac{\mathbf{S}_0^T \mathbf{S}_0}{\mathbf{S}_0^T A \mathbf{S}_0} = \frac{162}{810} = 0.2.$$



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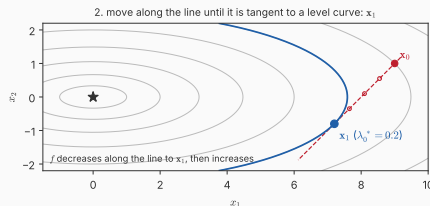
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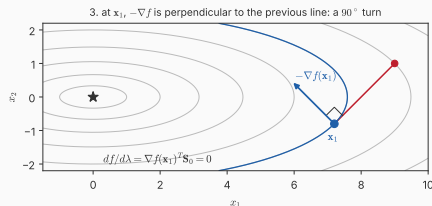
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Surprise: $\nabla f_1^T \mathbf{S}_0 = -64.8 + 64.8 = 0$.



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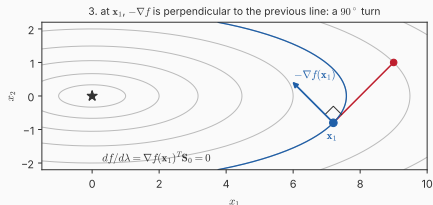
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Why: exact λ_k^*

At the minimum of $f(\mathbf{x}_k + \lambda \mathbf{S}_k)$: $\frac{df}{d\lambda} = \nabla f_{k+1}^T \mathbf{S}_k = 0$.

Since $\mathbf{S}_k = -\nabla f_k$: $\nabla f_{k+1}^T \nabla f_k = 0$

This holds for any differentiable f , provided λ_k^* minimizes f along $\mathbf{S}_k$ (it is not specific to quadratics): **consecutive gradients are orthogonal**. In 3.3.3 we will look for a better notion — conjugacy with respect to $H = \nabla^2 f$.

The algorithm

Steepest descent: $\mathbf{x}_0, \varepsilon$ (notation: $\nabla f_k = \nabla f(\mathbf{x}_k)$)

- Step 1** Choose the starting point $\mathbf{x}_0$ and $\varepsilon > 0$; set $k = 0$.
- Step 2** Search direction: set $\mathbf{S}_k = -\nabla f_k$.
- Step 3** Optimal step length: λ_k^* minimizes $f(\mathbf{x}_k + \lambda \mathbf{S}_k)$, $\lambda > 0$.
- Step 4** Set $\mathbf{x}_{k+1} = \mathbf{x}_k + \lambda_k^* \mathbf{S}_k$.
- Step 5** If $\|\nabla f_{k+1}\| \leq \varepsilon$, **stop**: $\mathbf{x}_{k+1}$ is the solution.
Otherwise, set $k = k + 1$ and go back to step 2.

How to compute λ_k^* (step 3)

- **1st method** (any f): substitute $\mathbf{x}_k + \lambda \mathbf{S}_k$ into f and solve $df/d\lambda = 0$; on the computer, the line search of Ch. 2.
- **2nd method** (quadratics only,

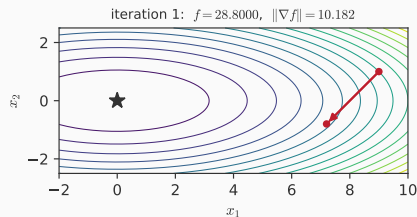
$$f = \frac{1}{2} \mathbf{x}^T \mathbf{A} \mathbf{x} + \mathbf{B} \mathbf{x} + C): \lambda_k^* = \frac{\mathbf{S}_k^T \mathbf{S}_k}{\mathbf{S}_k^T \mathbf{A} \mathbf{S}_k} \quad (\text{with } \mathbf{S}_k = -\nabla f_k).$$

- By hand, step 5 tests $\nabla f_{k+1} = \mathbf{0}$; on the computer, $\|\nabla f_{k+1}\| \leq \varepsilon$ and a maximum number of iterations $k_{\max}$.
- **Cost per iteration**: one gradient ($2n$ evaluations of f if numerical) + the line search. **Memory**: $O(n)$.
- **In practice**: *inexact* line search (sufficient decrease: Armijo/Wolfe).

The zigzag path

$f(\mathbf{x}) = \frac{1}{2}(x_1^2 + 9x_2^2)$ from $(9, 1)$, exact line search.

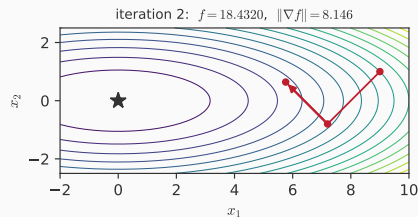
k	$\mathbf{x}_k$	$f(\mathbf{x}_k)$	$\ \nabla f_k\ $	λ_{k-1}^*
0	(9.0000, 1.0000)	45.00000	12.7279	—
1	(7.2000, -0.8000)	28.80000	10.1823	0.2000



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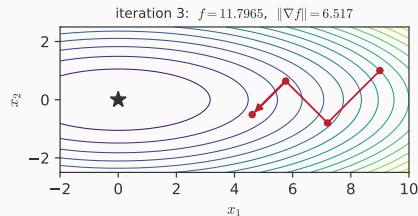
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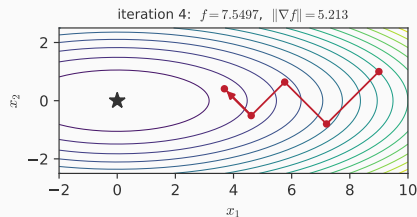
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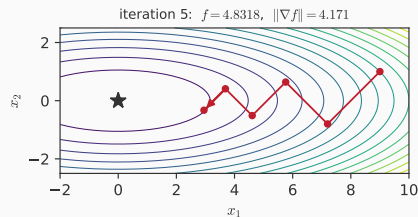
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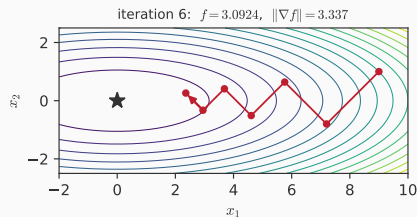
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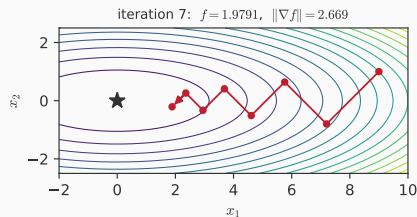
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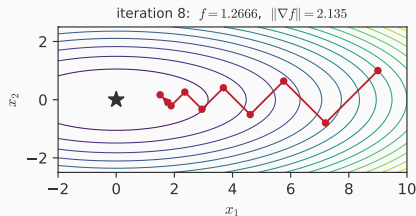


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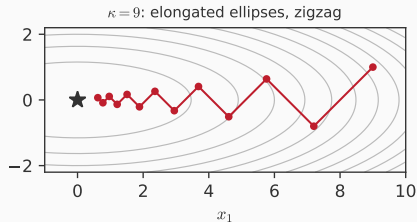
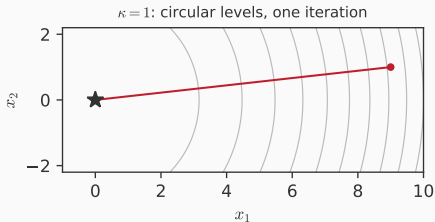
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8	(1.5099, 0.1678)	1.26664	2.1354	0.2000

Each iteration multiplies f by 0.64 and the step is always $\lambda^* = 0.2$: the path is geometrically similar from one iteration to the next. It takes **74 iterations** to reach $\|\nabla f\| < 10^{-6}$ — on a quadratic in two variables.



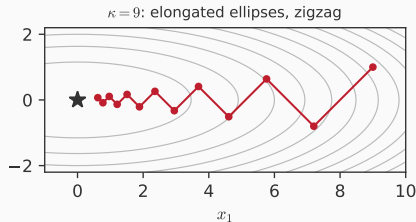
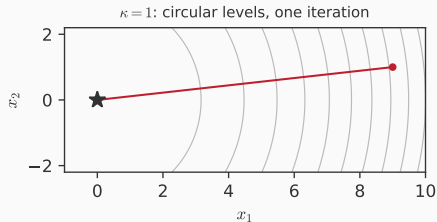
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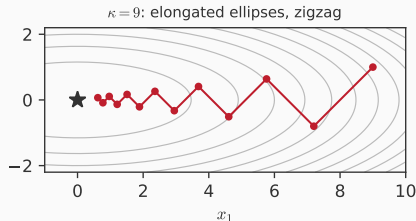
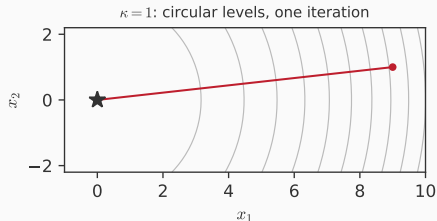
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Why, with circles, is one iteration enough? With circles ($\kappa = 1$), the normal to the level curve passes through the centre: $-\nabla f$ points **to the minimum**. With 3 : 1 ellipses ($\kappa = \lambda_{\max}/\lambda_{\min}$ of the Hessian: here 9/1; see next page), it points to the opposite “wall”, and the 90° steps go from wall to wall: 74 iterations.

Conditioning: why elongation = slowness

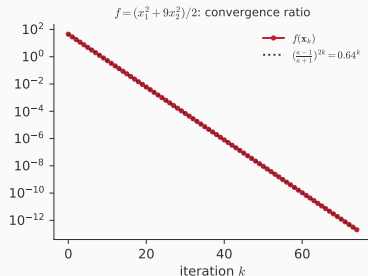
Assumptions: $f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T A \mathbf{x} + B\mathbf{x} + C$ with A **symmetric positive definite**; steepest descent with **exact** line search. Let $\mathbf{x}^*$ be the minimizer and $E(\mathbf{x}) = f(\mathbf{x}) - f^* = \frac{1}{2} \|\mathbf{x} - \mathbf{x}^*\|_A^2$, with $\|\mathbf{v}\|_A^2 = \mathbf{v}^T A \mathbf{v}$.

Bound (Kantorovich)

$$E(\mathbf{x}_{k+1}) \leq \left(\frac{\kappa - 1}{\kappa + 1} \right)^2 E(\mathbf{x}_k), \quad \kappa = \frac{\lambda_{\max}(A)}{\lambda_{\min}(A)},$$

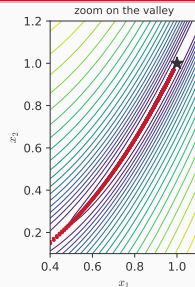
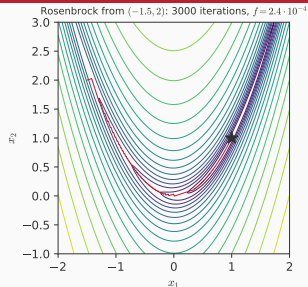
that is, $\|\mathbf{x}_{k+1} - \mathbf{x}^*\|_A \leq \frac{\kappa-1}{\kappa+1} \|\mathbf{x}_k - \mathbf{x}^*\|_A$: factor $\left(\frac{\kappa-1}{\kappa+1} \right)^2$ on the error **in f** , factor $\frac{\kappa-1}{\kappa+1}$ on the error **in the A -norm** ($\lambda_{\max}, \lambda_{\min}$: eigenvalues of A).

Here $\kappa = 9$, $f^* = 0$: the bound gives 0.64 for f , attained at every iteration from $(9, 1)$. $\kappa = 1$: one iteration; $\kappa = 100$: 0.96 — hundreds of iterations. **Linear** convergence, slow if κ is large.



It is a worst-case bound; for other quadratics there may be no equality at each iteration. The semi-axes of the level ellipses are in the ratio $\sqrt{\kappa}$.

Rosenbrock: reaching the valley is not the same as travelling along it



- It reaches the valley in a few iterations (the gradient is large and points there) and then **crawls**: after 3000 iterations still $f = 2.4 \cdot 10^{-4}$, $\mathbf{x} \approx (0.98, 0.97)$.
- At the minimum $(1, 1)$ the Hessian has $\kappa \approx 2.5 \cdot 10^3$: with exact line search, consecutive gradients are orthogonal and progress along the valley is very slow.
- Even so, it is one gradient and one line search per iteration: with very large n , first-order methods become attractive for their low cost and memory — it is the starting point of neural network training (“gradient descent”).

The core of one iteration, in MATLAB:

```
g = grad(x);           %  $n_g = 1$ 
S = -g;                % direction  $S_k$ 
phi = @(lam) f(x + lam*S); %  $f$  along  $S_k$ 
lam = fminbnd(phi, 0, lmax); %  $\lambda_k$  (Ch. 2)
% (if  $\lambda \sim lmax$ , enlarge  $lmax$ )
x = x + lam*S;
```

fminbnd uses golden section + parabolic interpolation (2.2–2.3).

- In the example ($\varepsilon_g = 10^{-6}$): 74 iterations, $n_g = 75$, about 6 evaluations of f per line search with this fminbnd (with the high-precision search of the figures, about 24).
- Without an analytical gradient, each gradient by central differences costs $n_f = 2n$ (2.6).
- The full functions (SteepestDescent, with info.nfev and info.ngev; GradFD) are in the course companion code.

The price of the steepest descent method

Advantages

- needs only ∇f (first order);
- cheap per iteration; memory $O(n)$;
- simple; it is the basis of many methods for very large n .

Limitations

- zigzag: consecutive gradients are orthogonal;
- linear convergence, very slow if κ is large;
- ignores curvature: in narrow valleys it points at the wall, not at the bottom.

“Global convergence” of a local algorithm

Global = from *any* starting point (under the assumptions: bounded level set, Lipschitz gradient (R), suitable line search) one obtains approximate *stationarity* ($\|\nabla f(\mathbf{x}_k)\| \rightarrow 0$, or at least $\liminf \|\nabla f(\mathbf{x}_k)\| = 0$, depending on the assumptions): the guarantee is stationarity, not necessarily convergence of the iterates to a single point. **It does not mean convergence to the global minimum** — in Ch. 5, “global” will refer to the optimum.

KEY TAKEAWAYS — 3.3.1

1. With only ∇f : $\mathbf{S}_k = -\nabla f_k$, the **locally steepest** descent; the step length λ_k^* minimizes $f(\mathbf{x}_k + \lambda \mathbf{S}_k)$ ($df/d\lambda = 0$ or the line search of Ch. 2).
2. With exact λ_k^* , $\nabla f_{k+1}^T \nabla f_k = 0$: consecutive gradients are **orthogonal** — the zigzag.
3. On SPD quadratics, with exact line search, the error in f satisfies $E_{k+1} \leq \left(\frac{\kappa-1}{\kappa+1}\right)^2 E_k$: **linear** convergence, which degrades as κ grows.
4. Low cost per iteration: one gradient and one line search; memory $O(n)$.
5. “Global convergence” = stationarity from any $\mathbf{x}_0$, under assumptions — **not** the global minimum.

Steepest descent uses only the **slope** and suffers from the geometry. Two questions:

Can we use the **curvature**? → Newton (3.3.2)

Can we use **memory** without paying for the Hessian? → conjugate gradients (3.3.3)

References Full list: *Course bibliography*, at the end of the course PDF.

- Nocedal, J., Wright, S. J., *Numerical Optimization*, 2nd ed., Springer, 2006 — Ch. 3.
- Fletcher, R., *Practical Methods of Optimization*, 2nd ed., Wiley, 1987.

Resources

- Companion code: `SteepestDescent` (`info.nfev`, `info.ngev`), `GradFD`. MATLAB: `fminbnd` (line search).